FAST National University of Computer and Emerging Sciences

Karachi Campus | Department of Computer Science

PROJECT PROPOSAL

Google Classroom Clone

A C++ OOP Console Application Simulating a Learning Management System

Course:	Object Oriented Programming — Lab Project
Instructor:	Sir Muhammad Khalid Khan
Section:	BAI-2B
Semester:	Spring 2026
Submission Date:	April 2026

Group Members

#	Name	Roll Number	Responsibility
1	Taimoor	25K-0096	Full OOP Implementation & Architecture
2	Zain Noman	25K-0146	Testing, Documentation & Presentation
3	Abdul Farid	25K-0029	Demo Data, UI Polish & Code Review

Department of Computer Science | FAST-NUCES Karachi | Spring 2026

1. Project Title

Google Classroom Clone A fully object-oriented C++ console application that simulates a Learning Management System (LMS) with role-based access for Teachers and Students across multiple classrooms, supporting stream posts, assignments, quizzes, recordings, and live meetings.

2. Group Members

#	Name	Roll Number	Tasks Assigned
1	Taimoor (Lead)	25K-0096	Full class hierarchy design, all module classes (Stream, Classwork, OnlineClass), Classroom & Platform, menus, polymorphism demo
2	Zain Noman	25K-0146	Testing, seed data scenarios, boundary checks, viva documentation
3	Abdul Farid	25K-0029	Final report write-up, presentation slides, code review & comments

3. Problem Statement

Managing academic operations, posting announcements, assigning and collecting work, tracking grades, scheduling online sessions, and maintaining student–teacher communication currently requires multiple disconnected tools. There is no single, unified console-based platform that models all of these interactions through clean OOP design.

This project addresses that gap by implementing a cohesive C++ application where every academic entity (User, Student, Teacher, Classroom, Assignment, Quiz, Recording, Meeting, Submission, GradeTracker) is represented as a well-structured class with appropriate inheritance, encapsulation, and polymorphism.

4. Proposed Solution

We propose to develop a Google Classroom Clone a multi-class C++ OOP console application. The system provides two interactive menus (Teacher and Student) that navigate through the three core tabs of each classroom: Stream, Classwork, and Online Class.

Key Modules

- Authentication Module: Platform-level login using ID/password with dynamic_cast-based role routing.
- Stream Module: Teachers post announcements; students post questions; comment threads on any post.
- Classwork Module: Full assignment and quiz lifecycle creation, submission, and grading.
- Online Class Module: Recording upload with metadata; meeting scheduling, live-start, and end controls.
- Grade Tracker: Per-student, per-class grade records with total computation.

- Polymorphism Demo: MathTeacher, ScienceTeacher, EnglishTeacher, CSTeacher each override conductSpecialActivity() and have subject-specific helper methods.

5. Technology Stack

Layer	Technology	Purpose
Language	C++17	Core application and all class implementations
Paradigm	Object-Oriented Programming	Encapsulation, inheritance, polymorphism, composition
Storage	Fixed-size arrays (no STL vectors)	All dynamic data held in bounded arrays with count variables
I/O	cin / cout (console)	Interactive menu-driven text interface
Compiler	g++ / MinGW (C++17)	Build and validation
Version Control	Git + GitHub	Collaboration and history tracking

6. OOP Concepts Implemented

OOP Concept	Where Implemented
Classes & Objects	Post, Assignment, Quiz, Recording, Meeting, Submission, GradeTracker, Stream, Classwork, OnlineClass, User, Student, Teacher, Classroom, Platform
Inheritance	Student and Teacher both inherit from User; MathTeacher, ScienceTeacher, EnglishTeacher, CSTeacher inherit from Teacher
Polymorphism	conductSpecialActivity() is virtual in Teacher and overridden in all four derived teacher classes; getRole() overridden in Student and Teacher
Encapsulation	Private/protected data members throughout; public getter methods; menu logic hidden inside Platform and Classroom
Composition	Classroom HAS-A Stream, Classwork, OnlineClass; Student HAS-A GradeTracker array
Dynamic Dispatch	dynamic_cast<Teacher*> and dynamic_cast<Student*> used after login() returns User*
Virtual Destructor	User base class has virtual ~User() to ensure correct cleanup of derived objects
Arrays (no vectors)	All collections use fixed-size arrays with MAX_* constants and manual count variables — no STL vector used anywhere

7. Project Timeline

Week	Milestone	Deliverable
Week 1	Proposal & Planning	Project proposal document

Week 2	Class hierarchy design	UML class diagram; User, Student, Teacher stubs
Week 3	Module classes	Stream, Classwork, OnlineClass complete
Week 4	Classroom & Platform	Full navigation menus working end-to-end
Week 5	Polymorphism & derived teachers	All four teacher subclasses; grade tracker
Week 6	Testing & seed data	Demo scenarios; boundary and edge-case tests
Week 7	Documentation & Submission	Final report, code comments, viva prep

8. Marking Scheme (Reference)

Component	Marks	Deadline
Proposal	1	Week 1
Code Submission	3	Before Viva
Project Report	2	Before Viva
Viva Voce	4	28 April 2026 (2B)
Total	**10**	—

9. Division of Work

Member	Roll No.	Tasks Assigned
Taimoor (Lead)	25K-0096	Class hierarchy design, all module classes, Classroom, Platform, teacher/student menus, polymorphism
Zain Noman	25K-0146	Seed demo data, boundary testing, edge-case handling, viva question prep
Abdul Farid	25K-0029	Final report, code comments/documentation, presentation slide deck

10. Expected Outcome

Upon completion, the Google Classroom Clone will deliver:

- A fully functional, multi-role C++ OOP console application.
- Compilable, zero-warning code under C++17 with g++.
- Demonstrated inheritance hierarchy: User → Student / Teacher → MathTeacher, ScienceTeacher, EnglishTeacher, CSTeacher.
- Runtime polymorphism via virtual functions with dynamic dispatch.
- Complete classroom workflow: post → assign → submit → grade → view grades.
- All data stored in fixed-size arrays — no STL vector used anywhere.

- Interactive menus covering all three classroom tabs for both Teacher and Student roles.

Note: All group members are aware of and understand every component of the project. Each member must be able to explain, modify, and extend their assigned code during the viva evaluation.

Student Representative Signature	Instructor Approval
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25K-0096 (Taimoor)	Sir Muhammad Khalid Khan